OM GIRI GOSWAMI

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Skills

C & C++ Programming
Intermediate

Hardware Prototyping & IoT (Arduino / ESP32)
Intermediate

Web Development (HTML / CSS / Netlify
Beginner - Intermediate

AI-Assisted Architecture ('Vibe Coding')
Proficient

Circuit Simulation & Tools
Intermediate

Interests

Robotics & Embedded Prototyping

Competitive Hackathons

Explore Open-Source Technologies

Tech-Community Engagement

Continuous Tech Exploration

Certifications

National Cadet Corps (NCC)

'A' Grade Certificate Holder. Demonstrated high discipline, teamwork, and leadership through rigorous training protocols.

Summary

"Driven B.Tech ECE undergraduate with a proven track record of bridging hardware prototyping and AI-assisted software development. Recognized for architecting the winning 'JEC-RESOLVE' ecosystem at Code Kumbh 2.0 and securing direct qualification for IIT Kanpur's TechKriti '26. Combines rigorous NCC discipline with active technical contributions across JLUG and MATRIX to deliver high-impact, scalable engineering solutions."

Experience

MATRIX.JEC
Public Relations & Marketing Team Member
JEC Jabalpur

- **Driving** strategic communication campaigns and public relations initiatives to engage a diverse collegiate audience.
- **Managing** promotional content and outreach strategies to translate complex technical events into accessible community engagement.

JEC Linux User's Group (JLUG JEC)
Technical Team Member

- **Collaborating** within an open-source technical ecosystem focused on Linux system administration and rapid software development.
- **Participating** in peer-to-peer technical knowledge sharing and fostering open-source culture among engineering undergraduates.

Education

Jabalpur Engineering College (JEC) | Jabalpur, MP
B.Tech in Electronics & Communication Engineering (ECE)
A GRADE
College
August 2025 - Present

Govt Excellence H S School | betul , MP
Class XII (Higher Secondary) — MP Board | 2024
81.2 %
School

Govt Excellence H S School | betul , MP
Class X (High School) — MP Board | 2022
90 %
School

Projects

JEC-RESOLVE (Smart Grievance Ecosystem)
A full-stack digital grievance ecosystem designed to streamline student-administrator communication and track issue resolution effectively.

Architected the technical framework focusing on scalability and user accessibility.

- **Secured** "Best in Open Innovation" at the Code Kumbh 2.0 Hackathon.
- **Earned** direct qualification for the final hackathon round at TechKriti '26 (IIT Kanpur), bypassing preliminary screening based on technical merit.

Kinetic Showdown (RC Car)
A custom 4-wheel drive robotic chassis engineered for high-speed obstacle navigation and maximum power-to-weight ratio at the INERTIA 2.0 event.

Engineered the structural foundation using high-torque DC motors and an L298N Dual H-Bridge.

- **Developed** responsive C++ firmware to translate HC-05 Bluetooth operator inputs into precise real-time motor control.
- **Validated** hardware-software integration for precision steering and agility under strict competitive time constraints.

Embedded Smart Security Lock
A hardware security prototype utilizing an Arduino, numeric matrix keypad, and servo motor actuation for controlled access.

Prototyped an embedded hardware security system for automated access control.

- **Validated** embedded C++ logic and circuitry performance through Wokwi schematic simulations prior to physical deployment.

Touchless IoT Safety System
A proximity-based automated safety prototype using an Arduino and ultrasonic sensors for non-contact operation.

Designed the sensor integration and logic workflows for automated, hygiene-focused operations.

- **Tested** and debugged hardware logic within Tinkercad simulation environments.